AI Incidents and Ethics: IBM Watson for Oncology

Marcelo Jimenez

Indiana Wesleyan University

AIML-500: AI and Machine Learning

Professor Cliff Birdsell

April 2026

Case Study Summary

In 2012, IBM initiated a collaboration with Memorial Sloan Kettering Cancer Center (MSK) — ranked the world's number one oncology hospital by Newsweek (2026) — to develop an artificial intelligence system capable of assisting oncologists in treatment decisions. Three years later, in 2015, IBM launched Watson for Oncology commercially. The system reached 230 hospitals across 14 countries — including India, China, South Korea, Thailand, and the United States — deploying it for clinical decisions on real patients (Ross & Swetlitz, 2018). By 2018, more than a dozen hospitals had canceled their contracts, citing unsafe and incorrect treatment recommendations inconsistent with local clinical guidelines, and high costs with no measurable clinical benefit (Dolfing, 2024; Ross & Swetlitz, 2018). In 2022, IBM sold Watson Health. The program was fully discontinued in 2023 (Dolfing, 2024).

Critical Analysis

Ethical Issue #1 — Synthetic Data as the Foundation of a Medical Decision System

Why I chose this issue:

This is the root error — the flaw that made all the others possible. An AI system trained on imaginary patients and deployed on real ones is not an implementation error: it is a fundamental design error. IBM shipped a product without real validation in a domain where an error is not measured in UX metrics — it is measured in lives.

Analysis:

IBM publicly stated that Watson had processed more than 600,000 pieces of medical evidence and 1.5 million real patient records from Memorial Sloan Kettering (Ross & Swetlitz, 2018). The reality, according to internal documents, was different: the training was built on hypothetical cases generated by just 1-2 specialists per cancer type, with no statistical input defining how many cases were sufficient (Ross & Swetlitz, 2018). The variables a physician manually entered were limited to tumor type, clinical stage, pathology report, and genetic mutations — entirely absent were active comorbidities, prior medications, allergy history, and regional context, the variables that in real clinical practice determine whether a treatment is viable (ScienceInsights, 2025). The system covered only 13 cancer types, in a field where the NIH Cancer Genome Atlas documents more than 200 recognized forms and 33 well-characterized primary types (NIH, 2018). Machine learning literature is precise on this point: synthetic data generated without statistical validation against a real population produces models that learn artificial patterns rather than the actual distribution of the problem. The bias present in the source data does not disappear — it transfers and amplifies in the resulting model (Veeramachaneni, MIT News, 2025; United Nations University, 2023). The model did not learn oncology: it learned the treatment preferences of 1-2 doctors in New York. At a cost of $200 to $1,000 USD per patient (Watson Health Management, 2018), hospitals paid validated-technology prices for a system whose training foundation was never what IBM claimed it was. That leaves a question no incident report answers: where was the ethics of the engineers and product managers who approved go-to-production for a system built on these foundations? That decision was not made by any algorithm.

Proposed Solution:

Prior to any deployment, require a statistically defined minimum of real cases per condition type, with the broadest possible dataset to capture real-world diversity — including comorbidities, demographic representation, and regional context — documented and audited by an independent third party. When training data does not represent the variability of the target population, the model fails silently on the cases it was never designed for (Ntoutsi et al., 2021).

The concept of the data nutrition label — proposed in AI ethics literature (Holland et al., 2018) — applies directly here: the origin, volume, and composition of the training set must be public and verifiable before go-live, equivalent to what the FDA requires for drugs prior to approval.

Ethical Issue #2 — Deliberate Misrepresentation of the System's Capabilities

Why I chose this issue:

What stands out is the human decision — not the algorithm's — to conceal that error while the product kept selling. A bug in production is an accident. Publicly declaring capabilities that internal documents directly contradicted is a decision. Selling a product that promises to do something it was never designed for, without technical backing, undermines trust in a tool whose end user is a physician — and that misplaced trust determines whether a patient lives or dies.

Analysis:

IBM's Senior VP publicly declared that Watson "has ingested all of Memorial Sloan Kettering's historical patient data" (Ross & Swetlitz, 2018). Internal documents confirmed this was false. Internal validation studies were deliberately designed to generate favorable results, not to measure the system's actual performance (Ross & Swetlitz, 2018). IBM never conducted prospective studies on real clinical impact, and actively lobbied for Watson to be exempt from FDA federal regulation — eliminating the only available external control mechanism (STAT News, 2018). The MD Anderson Cancer Center project was canceled after 4 years and $62.1 million in spending without treating a single patient (Dolfing, 2024). In 2023, the SEC fined IBM $2.3 million for misleading statements about Watson Health (Sweet TnT Magazine, 2025). Physicians adjusted their oversight level believing they were backed by 1.5 million real cases — a miscalibrated trust that the literature identifies as clinically dangerous (Cabitza et al., 2024). Academically, this pattern has a name: AI washing — the practice of exaggerating or fabricating

AI capabilities to attract investment or market share, creating information asymmetries between organizational claims and verifiable reality (Unver et al., 2026). IBM built a narrative it knew was false, sustained it while closing new contracts, and lobbied to eliminate the mechanism that would have exposed it (Ross & Swetlitz, 2018). That pattern is what turned a design error into documented human harm.

Proposed Solution:

Public declarations about an AI system's capabilities must be subject to the same verifiability standard the SEC requires for financial statements: anyone claiming their system was trained on X data must be able to demonstrate it to an independent auditor. Validation teams must be structurally separated from the commercial team — with no shared reporting line — to eliminate the incentive to design studies that confirm what one wants to sell. Just as accounting requires mandatory year-end audits and external reviews to prevent information asymmetries, technology — especially software with global reach affecting millions of people simultaneously — must have equivalent systems of continuous and mandatory auditing, with real regulatory consequences for those who evade them (Cabitza et al., 2024; Unver et al., 2026).

Ethical Issue #3 — Direct Patient Harm With Internal Knowledge

Why I chose this issue:

The core problem of this issue is that harm reached real patients without anyone being required to report it, track it, or correct it. This is the point where the entire chain of technical and commercial decisions becomes human consequence. IBM knew from day one it had a defective product. It did not know when it would explode. And when it did, no mandatory system existed to register the damage.

Analysis:

As of July 2017, internal IBM presentations documented "multiple examples of unsafe and incorrect treatment recommendations" (Ross & Swetlitz, 2018). The most concrete and documented case: a 65-year-old patient with lung cancer and active severe bleeding received a recommendation for bevacizumab — a drug with an FDA black box warning explicitly contraindicating its use in patients with severe bleeding due to risk of fatal hemorrhage (Advisory Board, 2018). The recommendations were inconsistent with National Comprehensive Cancer Network (NCCN) guidelines, the reference standard in oncology (Ross & Swetlitz, 2018). During that same period, IBM's sales team continued closing contracts with new hospitals. That same exposed population — 84,000 patients across 230 hospitals — never had access to a mechanism that would register whether the system had contributed to harm in their treatment (Advisory Board, 2018). No adverse events were publicly reported — which does not mean they did not occur: it means no mandatory reporting and traceability mechanisms existed for health AI systems. To date, no equivalent federal mechanism has been established for health AI systems.

Proposed Solution:

Once a technology system that directly affects the life or health of people is in production, a federal adverse event reporting mechanism equivalent to the FDA's FAERS system for drugs must exist: each documented failure generates a report number, a public registry, and auditable traceability. If a drug causes harm, that number exists. If an algorithm causes harm today, the equivalent does not. Additionally, hospitals and companies deploying these systems must have a legal obligation to notify exposed patients when systemic failures are documented

— the same disclosure principle that applies to personal data breaches under regulations such as HIPAA (Cabitza et al., 2024; Ross & Swetlitz, 2018).

Reflection on Critical Thinking

I came into this analysis thinking the core problem was technical — insufficient testing and validation, a rushed deployment, and possibly inadequate end-user training.

What changed was understanding that this is not a garbage in, garbage out case — that explanation becomes insignificant against the real magnitude of the problem. This case accumulated human errors from day one. The most basic Software Development Life Cycle (SDLC) was not followed: requirements analysis, design, implementation, testing, deployment, and maintenance. Every stage was loaded with incorrect decisions that, like a snowball, kept growing until they exploded — and when they did, they exploded on real patients.

When an AI system causes the death of one person, the harm has a name and the legal system compels accountability. When it destroys thousands of people, the harm disperses across thousands of families and their futures — a mortgage at risk, a family relocated, a child's future compromised — stories that never appear in the financial report.

Is harm less serious because it is difficult to measure?

Disclosure

Claude-Sonnet 4.6 was used to support research and source identification for this case study. The critical analysis, arguments, and reflections are entirely my own.

References

Advisory Board. (2018, July 27). *IBM Watson recommended 'unsafe and incorrect' treatments for cancer patients, investigation reveals.* https://www.advisory.com/daily-briefing/2018/07/27/ibm

AI Incident Database. (n.d.). *Incident #225: IBM Watson for Oncology criticized for unsafe and inaccurate cancer treatment recommendations.* https://incidentdatabase.ai/cite/225/

Cabitza, F., Campagner, A., & Natali, C. (2024). The unmet promise of trustworthy AI in healthcare: Why we fail at clinical translation. *Frontiers in Digital Health.* https://pmc.ncbi.nlm.nih.gov/articles/PMC11063331/

Dolfing, H. (2024). *Case study 20: The $4 billion AI failure of IBM Watson for Oncology*. https://www.henricodolfing.com/2024/12/case-study-ibm-watson-for-oncology-failure.html

Holland, S., Hosny, A., Newman, S., Joseph, J., & Chmielinski, K. (2018). *The dataset nutrition label: A framework to drive higher data quality standards.* arXiv. https://arxiv.org/abs/1805.03677

Memorial Sloan Kettering Cancer Center. (2025). *MSK earns U.S. News "best hospital" ranking for 36 consecutive years.* https://www.mskcc.org/u-s-news-world-report-names-msk-top-hospital-cancer-care

Newsweek. (2026). *World's best specialized hospitals 2026: Memorial Sloan Kettering Cancer Center ranked #1 in oncology.* https://www.mskcc.org/news-releases/mskcc-is-best-oncology-hospital-in-world-according-to-newsweek

NIH National Cancer Institute. (2018). *Genomic analysis of 33 cancer types completed.* https://www.nih.gov/news-events/nih-research-matters/genomic-analysis-33-cancer-types-completed

Ntoutsi, E., Fafalios, P., Gadiraju, U., Iosifidis, V., Nejdl, W., Vidal, M. E., & Staab, S. (2021). Bias in data-driven artificial intelligence systems — An introductory survey. *WIREs Data Mining and Knowledge Discovery, 10(3).* https://pmc.ncbi.nlm.nih.gov/articles/PMC8515002/

Ross, C., & Swetlitz, I. (2018, July 25). *IBM's Watson supercomputer recommended unsafe and incorrect cancer treatments, internal documents show.* STAT News. https://www.statnews.com/2018/07/25/ibm-watson-recommended-unsafe-incorrect-treatments/

ScienceInsights. (2025, November 23). *What happened to IBM Watson Oncology?* https://scienceinsights.org/what-happened-to-ibm-watson-oncology/

Sweet TnT Magazine. (2025). *AI washing: The billion-dollar scam sabotaging investors and tech innovation.* https://sweettntmagazine.com/ai-washing-hidden-financial-risks-investors/

United Nations University. (2023). *The use of synthetic data to train AI models: Opportunities and risks for sustainable development.* https://unu.edu/publication/use-synthetic-data-train-ai-models-opportunities-and-risks-sustainable-development

Unver, B., & Asan, O. (2026). *AI washing and the erosion of digital legitimacy: A socio-technical perspective on responsible artificial intelligence in business.* arXiv. https://arxiv.org/html/2601.06611

Veeramachaneni, K. (2025, September 3). 3 questions: The pros and cons of synthetic data in AI. *MIT News.* https://news.mit.edu/2025/3-questions-pros-cons-synthetic-data-ai-kalyan-veeramachaneni-0903

Watson Health Management. (2018). IBM Watson Health: Managing partnerships to improve personalized medicine. *eCampus Ontario Press.* https://ecampusontario.pressbooks.pub/bio16610w18/chapter/ibm-watson-health-managing-partnerships-to-improve-personalized-medicine/